

Problem 1

Solution:

$$A = \{1, -1\}$$

$$B = \{1\}$$

$$C = \{1\}$$

$$D = \{2\}$$

$$E = \{1, -17\}$$

$$F = \{1, -17, -8 + \sqrt{47}, -8 - \sqrt{47}\}$$

□

Problem 2

Solution:

Every set is a subset of itself. Omitting those statements for concision.

$$B \subseteq C$$

$$C \subseteq B$$

$$B \subseteq A$$

$$C \subseteq A$$

$$B \subseteq E \subseteq F$$

$$C \subseteq E \subseteq F$$

□

Problem 3

Solution:

(a) True

(b) True

(c) False

(d) True

(e) False

(f) False

□

Problem 4

Solution:

- (a) True
- (b) True
- (c) True
- (d) True
- (e) False
- (f) False

□

Problem 5

Solution:

$$\begin{aligned}\mathcal{P}(S) = \{ & \\ & \emptyset, \\ & \{1\}, \\ & \{2\}, \\ & \{3\}, \\ & \{4\}, \\ & \{1, 2\}, \\ & \{1, 3\}, \\ & \{1, 4\}, \\ & \{2, 3\}, \\ & \{2, 4\}, \\ & \{3, 4\}, \\ & \{1, 2, 3\}, \\ & \{1, 2, 4\}, \\ & \{1, 3, 4\}, \\ & \{2, 3, 4\}, \\ & S\end{aligned}$$

□

Problem 6

Solution:

- (a) False
- (b) False
- (c) False
- (d) True
- (e) False
- (f) False
- (g) True
- (h) False
- (i) True

□

Problem 7

Solution:

(a) Let $x \in \{a, a\}$. Then, $x = a$ which implies $x \in \{a\}$. As a result, we can say $\{a, a\} \subseteq \{a\}$. By the exact same logic, we can also say $\{a\} \subseteq \{a, a\}$. Thus, $\{a, a\} = \{a\}$.

(b) Let $x \in \{a, b\}$. Then, $x = a$ or $x = b$. In either case $x \in \{b, a\}$. As a result, we can say $\{a, b\} \subseteq \{b, a\}$. By the exact same logic, we can also say $\{b, a\} \subseteq \{a, b\}$. Thus, $\{a, b\} = \{b, a\}$.

(c) Suppose $\{a\} = \{b, c\}$. Then, $\{b, c\} \subseteq \{a\}$. In other words if $x \in \{b, c\}$, then $x \in \{a\}$. There are two cases ($x = b$ or $x = c$) but in either case we must have $x = a$. Therefore, $b = c = a$.

Now let's show the reverse case. Suppose $a = b = c$. Then, $x \in \{a\} \implies x \in \{b, c\}$ and $x \in \{b, c\} \implies x \in \{a\}$. In other words $\{a\} \subseteq \{b, c\}$ and $\{b, c\} \subseteq \{a\}$. But that means that $\{a\} = \{b, c\}$, as desired. □

Problem 8

Solution:

$$x \in A \cup B \implies (x \in A) \vee (x \in B) \implies (x \in B) \vee (x \in A) \implies x \in B \cup A$$

so $A \cup B \subseteq B \cup A$. Similarly,

$$x \in B \cup A \implies (x \in B) \vee (x \in A) \implies (x \in A) \vee (x \in B) \implies x \in A \cup B$$

so $B \cup A \subseteq A \cup B$. And now, we can conclude $A \cup B = B \cup A$.

The proof of $A \cap B = B \cap A$ is identical, so we omit it.

□

Problem 9

Solution:

$$\begin{aligned} x \in A \cup (B \cup C) &\implies x \in A \vee x \in (B \cup C) \\ &\implies x \in A \vee (x \in B \vee x \in C) \\ &\implies x \in A \vee x \in B \vee x \in C \\ &\implies (x \in A \vee x \in B) \vee x \in C \\ &\implies x \in (A \cup B) \cup C \end{aligned}$$

so $A \cup (B \cup C) \subseteq (A \cup B) \cup C$

$$\begin{aligned} x \in (A \cup B) \cup C &\implies (x \in A \vee x \in B) \vee x \in C \\ &\implies (x \in A \vee x \in B) \vee x \in C \\ &\implies x \in A \vee x \in B \vee x \in C \\ &\implies x \in A \vee (x \in B \vee x \in C) \\ &\implies x \in A \cup (B \cup C) \end{aligned}$$

so $(A \cup B) \cup C \subseteq A \cup (B \cup C)$ and we can conclude $A \cup (B \cup C) = (A \cup B) \cup C$

The proof of $A \cap (B \cap C) = (A \cap B) \cap C$ is identical, so we omit it.

□

Problem 10

Solution:

$$\begin{aligned}
 x \in A \cap (B \cup C) &\implies x \in A \wedge x \in (B \cup C) \\
 &\implies x \in A \wedge (x \in B \vee x \in C) \\
 &\implies (x \in A \wedge x \in B) \vee (x \in A \wedge x \in C) \\
 &\implies (x \in A \cap B) \vee (x \in A \cap C) \\
 &\implies x \in (A \cap B) \cup (A \cap C)
 \end{aligned}$$

so $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$

$$\begin{aligned}
 x \in (A \cap B) \cup (A \cap C) &\implies (x \in A \wedge x \in B) \vee x \in A \cap C \\
 &\implies (x \in A \wedge x \in B) \vee (x \in A \wedge x \in C) \\
 &\implies x \in A \wedge (x \in B \vee x \in C) \\
 &\implies x \in A \wedge (x \in B \cup C) \\
 &\implies x \in A \cap (B \cup C)
 \end{aligned}$$

so $(A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C)$ and we can conclude $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

The proof of $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ is identical, so we omit it.

□

Problem 11

Solution:

$$x \in A \implies x \in A \vee x \in A \implies x \in A \cup A \implies A \subseteq A \cup A$$

$$x \in A \cup A \implies x \in A \vee x \in A \implies x \in A \implies A \cup A \subseteq A$$

Thus, we can conclude that $A \cup A = A$. The proof of $A \cap A = A$ is identical, so we omit it.

□

Problem 12

Solution:

$$x \in A \implies x \in A \vee x \in B \implies x \in A \cup B \implies A \subseteq A \cup B$$

$$x \in A \cap B \implies x \in A \wedge x \in B \implies x \in A \implies A \cap B \subseteq A$$

□

Problem 13

Solution:

$$x \in A \cup \emptyset \implies x \in A \vee x \in \emptyset \implies x \in A \text{ (since } x \in \emptyset \text{ identically false)} \implies A \cup \emptyset \subseteq A$$

$$\begin{aligned} x \in A &\implies x \in A \vee x \in \emptyset \text{ (works since we're assuming the first predicate is true)} \\ &\implies x \in A \cup \emptyset \\ &\implies A \subseteq A \cup \emptyset \end{aligned}$$

so we can conclude that $A \cup \emptyset = A$

For the second assertion,

$$\begin{aligned} x \in A \cap \emptyset &\implies x \in A \wedge x \in \emptyset \\ &\implies x \in \emptyset \\ &\implies A \cap \emptyset \subseteq \emptyset \end{aligned}$$

$$\begin{aligned} x \in \emptyset &\implies x \in A \wedge x \in \emptyset \text{ (vacuously true since } x \in \emptyset \text{ is identically false)} \\ &\implies x \in A \cap \emptyset \\ &\implies \emptyset \subseteq A \cap \emptyset \end{aligned}$$

so we can conclude that $A \cap \emptyset = \emptyset$

□

Problem 14

Solution:

$$\begin{aligned} x \in A \cup (A \cap B) &\implies x \in A \vee x \in (A \cap B) \\ &\implies x \in A \vee (x \in A \wedge x \in B) \\ &\implies x \in A \vee x \in A \\ &\implies x \in A \\ &\implies A \cup (A \cap B) \subseteq A \end{aligned}$$

$$\begin{aligned} x \in A &\implies x \in A \vee (x \in A \wedge x \in B) \\ &\implies x \in A \vee x \in (A \cap B) \\ &\implies x \in A \cup (A \cap B) \\ &\implies A \subseteq A \cup (A \cap B) \end{aligned}$$

so we can conclude that $A \cup (A \cap B) = A$

A brief note on the proof strategy in the reverse case: any expression works after the $\vee$ in the first implication statement (because we're assuming the first part is true). In other words, $X \vee Y$ is always true if we know X is true. We just chose $x \in A \wedge x \in B$ because that was useful to what we were trying to show

For the second assertion,

$$\begin{aligned} A \cap (A \cup B) &= (A \cap A) \cup (A \cap B) && \text{(by the distributive law)} \\ &= A \cup (A \cap B) && \text{(by the reflexive property from problem 11)} \\ &= A && \text{(by the assertion we just proved in this problem)} \end{aligned}$$

□

Problem 15

Solution:

$$\begin{aligned} x \in A \cup B &\implies x \in A \vee x \in B \\ &\implies x \in C \vee x \in B && \text{since } A \subseteq C \\ &\implies x \in C \vee x \in C && \text{since } B \subseteq C \\ &\implies x \in C \\ &\implies A \cup B \subseteq C \end{aligned}$$

□

Problem 16

Solution:

$$\begin{aligned} x \in C &\implies x \in C \wedge x \in C \\ &\implies x \in A \wedge x \in C && \text{since } C \subseteq A \\ &\implies x \in A \wedge x \in B && \text{since } C \subseteq B \\ &\implies x \in A \cap B \\ &\implies C \subseteq (A \cap B) \end{aligned}$$

□

Problem 17

Solution:

(b) We'll solve this one first since we'll use the result in our proof of (a).

$$\begin{aligned} x \in A &\implies x \in B && \text{because } A \subseteq B \\ &\implies x \in C && \text{because } B \subseteq C \\ &\implies A \subseteq C \end{aligned}$$

(a) Recall that $A \subset B$ means that $A \subseteq B$ but $A \neq B$

$$\begin{aligned} x \in A &\implies x \in B && \text{because } A \subset B \implies A \subseteq B \\ &\implies x \in C && \text{because } B \subset C \implies B \subseteq C \\ &\implies A \subseteq C \end{aligned}$$

Now we have to show that $A \neq C$. We'll do it by contradiction. Suppose that $A = C$. Then, $C \subseteq A$, but this, in addition to the fact that $A \subset B \implies A \subseteq B$, means that $C \subseteq B$ as well by the assertion we just proved in (b). But if $C \subseteq B$ and $B \subset C \implies B \subseteq C$, then $B = C$. Contradiction! Because $B \subset C \implies B \neq C$

(c) $A \subset C$

(d) Yes

(e) No. Consider $x = 1$, $A = \{1\}$, and $B = \{\{1\}\}$. $x \in A$ and $A \in B$, but $x \notin B$. □

Problem 18

Solution:

$$\begin{aligned} x \in A - (B \cap C) &\implies x \in A \wedge x \notin (B \cap C) \\ &\implies x \in A \wedge \neg(x \in B \wedge x \in C) \\ &\implies x \in A \wedge (x \notin B \vee x \notin C) \\ &\implies (x \in A \wedge x \notin B) \vee (x \in A \wedge x \notin C) \\ &\implies x \in A - B \vee x \in A - C \\ &\implies x \in (A - B) \cup (A - C) \\ &\implies A - (B \cap C) \subseteq (A - B) \cup (A - C) \end{aligned}$$

$$\begin{aligned}
 x \in (A - B) \cup (A - C) &\implies (x \in A \wedge x \notin B) \vee (x \in A \wedge x \notin C) \\
 &\implies x \in A \wedge (x \notin B \vee x \notin C) \\
 &\implies x \in A \wedge \neg(x \in B \wedge x \in C) \\
 &\implies x \in A \wedge x \notin (B \cap C) \\
 &\implies x \in A - (B \cap C) \\
 &\implies (A - B) \cup (A - C) \subseteq A - (B \cap C)
 \end{aligned}$$

so we can conclude that $A - (B \cap C) = (A - B) \cup (A - C)$

□

Problem 19

Solution:

$$\begin{aligned}
 x \in B - \bigcup_{A \in \mathcal{F}} A &\implies x \in B \wedge x \notin \bigcup_{A \in \mathcal{F}} A \\
 &\implies x \in B \wedge \bigwedge_{A \in \mathcal{F}} (x \notin A) \\
 &\implies x \in B \wedge \bigwedge_{A \in \mathcal{F}} (x \in B \wedge x \notin A) \\
 &\implies \bigwedge_{A \in \mathcal{F}} (x \in B \wedge x \notin A) \\
 &\implies x \in \bigcap_{A \in \mathcal{F}} (B - A) \\
 &\implies B - \bigcup_{A \in \mathcal{F}} A \subseteq \bigcap_{A \in \mathcal{F}} (B - A)
 \end{aligned}$$

$$\begin{aligned}
x \in \bigcap_{A \in \mathcal{F}} (B - A) &\implies \bigwedge_{A \in \mathcal{F}} (x \in B - A) \\
&\implies \bigwedge_{A \in \mathcal{F}} (x \in B \wedge x \notin A) \\
&\implies x \in B \wedge \bigwedge_{A \in \mathcal{F}} (x \notin A) \\
&\implies x \in B \wedge \neg \left(\bigvee_{A \in \mathcal{F}} x \in A \right) \\
&\implies x \in B \wedge x \notin \bigcup_{A \in \mathcal{F}} A \\
&\implies x \in B - \bigcup_{A \in \mathcal{F}} A \\
&\implies \bigcap_{A \in \mathcal{F}} (B - A) \subseteq B - \bigcup_{A \in \mathcal{F}} A
\end{aligned}$$

so we can conclude

$$B - \bigcup_{A \in \mathcal{F}} A = \bigcap_{A \in \mathcal{F}} (B - A)$$

The proof of the other assertion follows an identical approach so we omit it for clarity. $\square$

Problem 20

Solution:

(a) (i) is sometimes incorrect. Let $A = \{1\}$, $B = \{1, 2\}$, and $C = \{1, 3\}$. Then $A - (B - C) = \{1\} - \{2\} = \{1\}$. However, $(A - B) \cup C = \emptyset \cup C = C = \{1, 3\}$ by the proposition from problem 13.

(ii) is always true.

$$\begin{aligned}
x \in (A - B) - C &\implies (x \in A - B) \wedge x \notin C \\
&\implies (x \in A \wedge x \notin B) \wedge x \notin C \\
&\implies x \in A \wedge (x \notin B \wedge x \notin C) \\
&\implies x \in A \wedge \neg(x \in B \vee x \in C) \\
&\implies x \in A \wedge \neg(x \in B \cup C) \\
&\implies x \in A - (B \cup C) \\
&\implies (A - B) - C \subseteq A - (B \cup C)
\end{aligned}$$

$$\begin{aligned}x \in A - (B \cup C) &\implies x \in A \wedge x \notin (B \cup C) \\&\implies x \in A \wedge \neg(x \in B \vee x \in C) \\&\implies x \in A \wedge (x \notin B \wedge x \notin C) \\&\implies (x \in A \wedge x \notin B) \wedge x \notin C \\&\implies x \in A - B \wedge x \notin C \\&\implies x \in (A - B) - C \\&\implies A - (B \cup C) \subseteq (A - B) - C\end{aligned}$$

so we can conclude $A - (B \cup C) = (A - B) - C$.

(b) $C = \emptyset$ or $C \subseteq A \iff A - (B \cup C) = (A - B) - C$.

□